



COURSE SLIDES · WSQ

Autodesk Certified Professional (ACP) - AutoCAD Design and Drafting

WSQ Course Code: TGS-2023036004

Conducted by Tertiary Infotech Academy Pte Ltd · UEN
201200696W

v8.0 · 22 August 2026

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COURSE ADMINISTRATION

Welcome & Course Orientation

Digital Attendance — Three Required Check-ins

1

AM attendance

Scan the SSG QR code at the start of each training day.

2

PM attendance

Scan again after lunch when prompted by the trainer.

3

Assessment attendance

Complete the separate assessment check-in before the final papers.

4

Eligibility

Maintain at least 75% attendance and complete all prescribed assessments.

About the Trainer



Your Trainer

General trainer profile template

NAME

DESIGNATION

QUALIFICATIONS

AUTOCAD EXPERTISE

INDUSTRY EXPERIENCE

CONTACT

About the Trainer



Dr Alfred Ang

Principal Trainer and Courseware
Developer
Tertiary Infotech Academy

ROLE

Principal trainer and courseware developer at Tertiary Infotech Academy.

QUALIFICATIONS

Doctorate-qualified adult educator with ACTA/DACE training credentials.

FOCUS

Professional software workflows, design communication, automation and emerging technologies.

APPROACH

Connect design intent to measurable standards, practice evidence and stakeholder outcomes.

Let's Know Each Other

1

Role

What kinds of drawings do you create or review?

2

Experience

Which AutoCAD workflows are already familiar?

3

Challenge

Where do revisions, standards or coordination currently fail?

4

Goal

What professional outcome should this course unlock?

Ground Rules

1

Set mobile devices to silent mode.

2

Participate actively — no question is too small.

3

Respect different professional viewpoints.

4

Use only authorised project files and data.

5

Return from breaks on time.

6

Keep evidence of your own work.

Download Your Course Materials

1

Open the LMS/TMS
course page

2

Locate the current
Courseware section

3

Download slides,
Learner Guide and labs

4

Open each lab from its
own folder

Four-Day Learning Journey

Days 1–2

- Day 1 — Standards and drawing management
 - Units, UCS, layers, properties and advanced objects
 - Labs 1–2
- Day 2 — Author, edit and annotate
 - Precision editing, blocks, attributes and annotation systems
 - Labs 3–6

Days 3–4

- Day 3 — Output and collaboration
 - Layouts, page setups, publish, eTransmit and Xrefs
 - Labs 7–9
- Day 4 — Integrate and demonstrate
 - Sink documentation capstone, evidence review, WA and PP
 - Lab 10 + final assessment

TSC Coverage — Knowledge and Abilities

Knowledge

- K1 · Emerging design visualization standards
- K2 · Revision methods for 2D and 3D drawings
- K3 · Storytelling and stakeholder engagement
- K4 · Emerging drawing technologies

Abilities

- A1 · Define art direction and sketching style
- A2 · Guide drawing capability development
- A3 · Develop organisational drawing standards
- A4 · Lead technology and method selection
- A5 · Inspire traditional and contemporary approaches

Learning Outcomes

1

LO1

Standards and art direction

2

LO2

Drawing and revision capability

3

LO3

Standards, storytelling and engagement

4

LO4

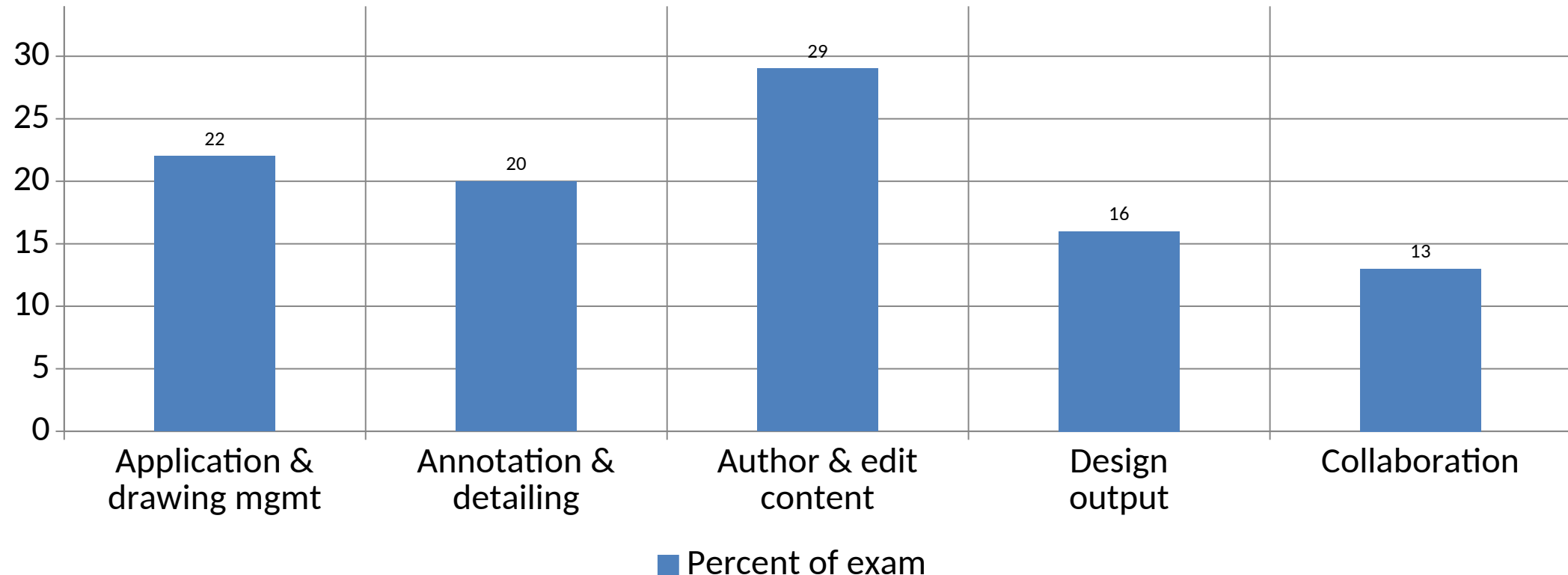
Emerging technology integration

5

LO5

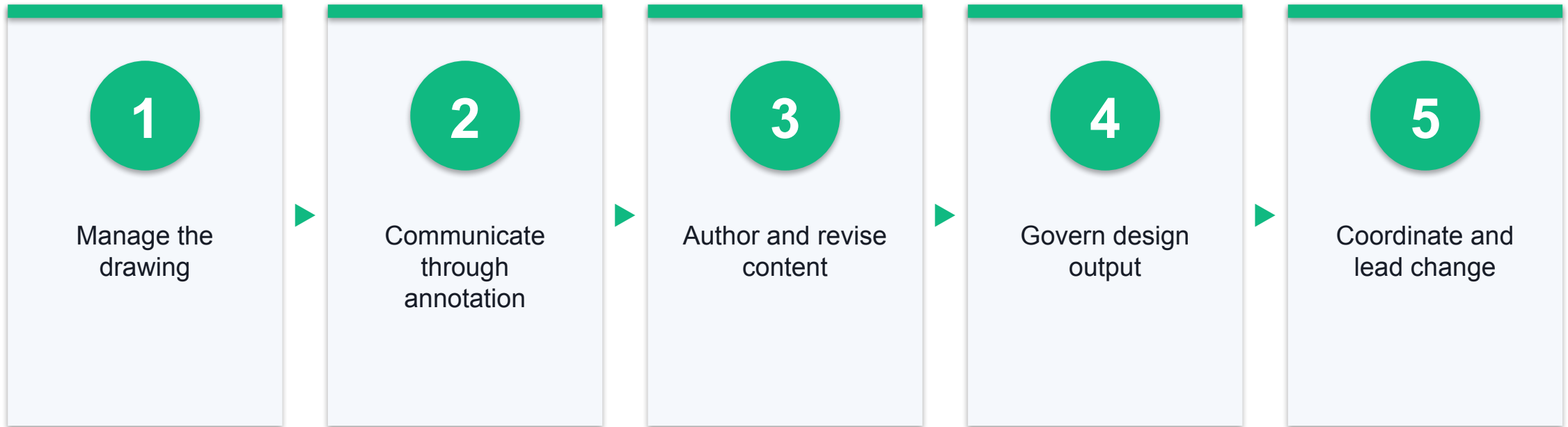
Exploration and stakeholder inspiration

Autodesk ACP Exam-Domain Weighting



Official Autodesk AutoCAD ACP domain weighting. Use it to balance review effort; WSQ assessment remains K1–K4 and A1–A5.

Course Roadmap



Briefing for Assessment

1

Individual, open-book assessment using approved course materials.

2

Place phones and other materials under the table.

3

No discussion, photos or recording of assessment scripts.

4

Use black/blue pen; no liquid paper or correction tape.

5

Answer in your own words and submit through LMS/TMS.

6

C requires competence in every required criterion.

Final Assessment

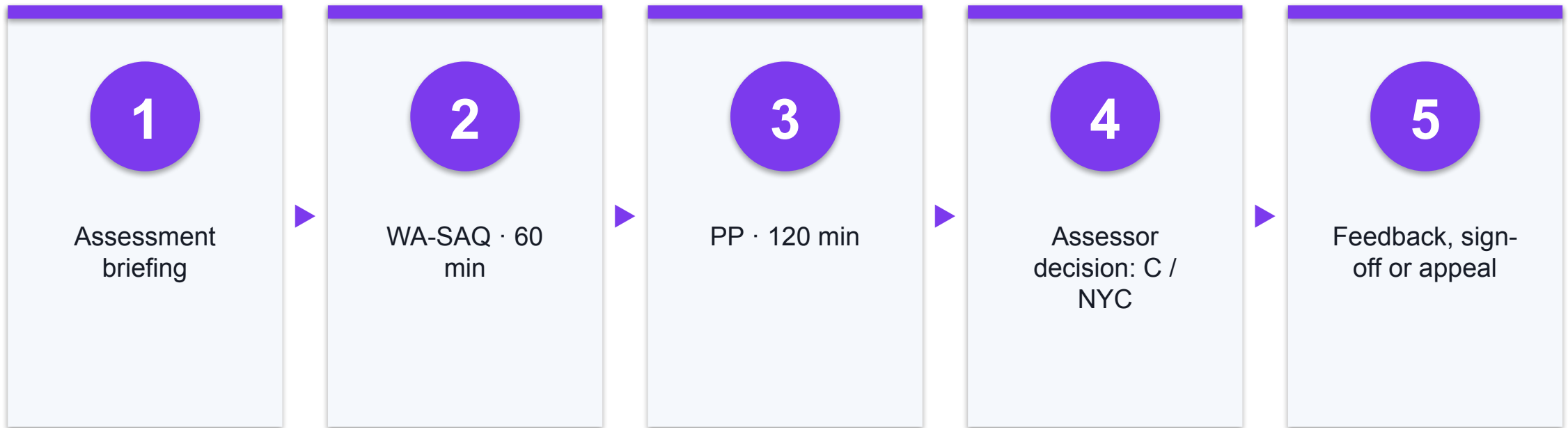
Written Assessment

- Written Assessment (WA-SAQ) — 4 open-ended questions covering K1–K4, 60 minutes, open book.
- Knowledge evidence from slides and Learner Guide.
- Four questions retain the original WA count.

Practical Performance

- Practical Performance (PP) — 5 hands-on tasks covering A1–A5, 120 minutes, open book.
- Practical evidence from the sink layout workflow.
- Five tasks retain the original PP count and A1–A5 mapping.

Assessment and Sign-off Flow



DOMAIN 01

Application and Drawing Management

Standards · units · UCS · layers · properties · drawing health

DRAWING GOVERNANCE

A reliable drawing begins with declared rules.

Units, origin, naming, layers and revision status are part of the design — not afterthoughts.

Project Setup — Six Controls

1

Purpose

What decision or deliverable must the drawing support?

2

Units

Length, precision, insertion scale and angle convention.

3

Origin

World coordinates plus named project UCS where required.

4

Layers

Names, properties, plot behaviour and ownership.

5

Objects

Fit-for-purpose types with measurable properties.

6

Revision

Status, filename, change record and evidence.

Units, Scale and Precision

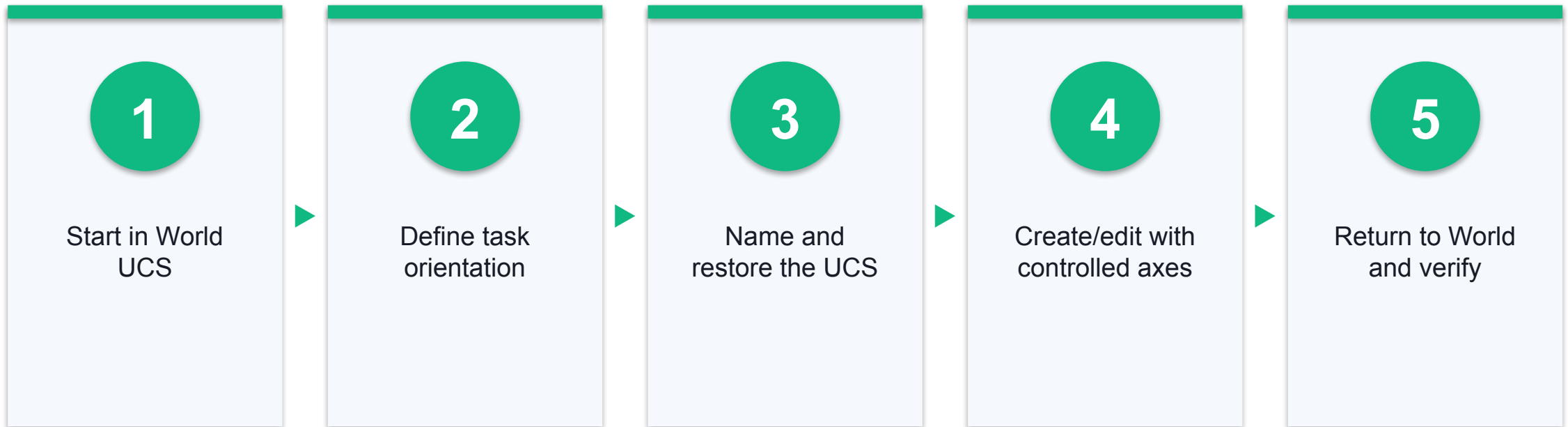
What each controls

- Drawing units describe model-space values.
- Insertion scale controls how referenced content enters.
- Viewport scale controls model-to-paper presentation.

Professional checks

- Check a known dimension before editing.
- Never fix a wrong model scale with a plot-scale workaround.
- Record precision appropriate to the decision.

User Coordinate Systems



Layer Governance

1

Properties

Colour, linetype, linewidth, transparency and plot style.

2

Visibility

Off for temporary visibility; Freeze for regeneration/performance.

3

Protection

Lock prevents accidental edits while keeping context visible.

4

Ownership

Xref prefixes and layer naming reveal source and responsibility.

5

Filters

Property and group filters focus complex drawings.

6

States

Named combinations restore review, plot or coordination views.

Layer States vs Viewport Overrides

Layer state

- Layer states capture a reusable visibility/property condition.
- Use them for named working, coordination and presentation modes.
- Restore globally or selectively.

Viewport override

- Viewport overrides change presentation in one layout viewport.
- Use them to emphasise sheet-specific information.
- Keep global layer properties intact.

Object Queries and Selection

1

Properties palette

Read and edit complete object properties.

2

Quick Select

Filter by object type and property.

3

Select Similar

Reuse the current object's property pattern.

4

Filter

Build compound selection criteria.

5

LIST / ID / DIST

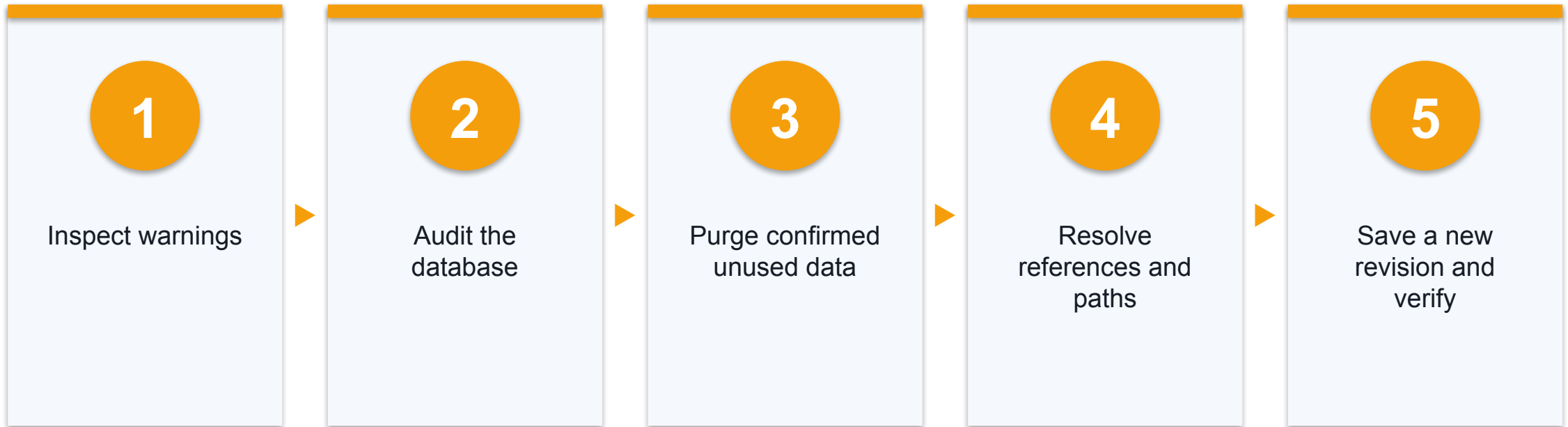
Verify coordinates, geometry and measurement.

6

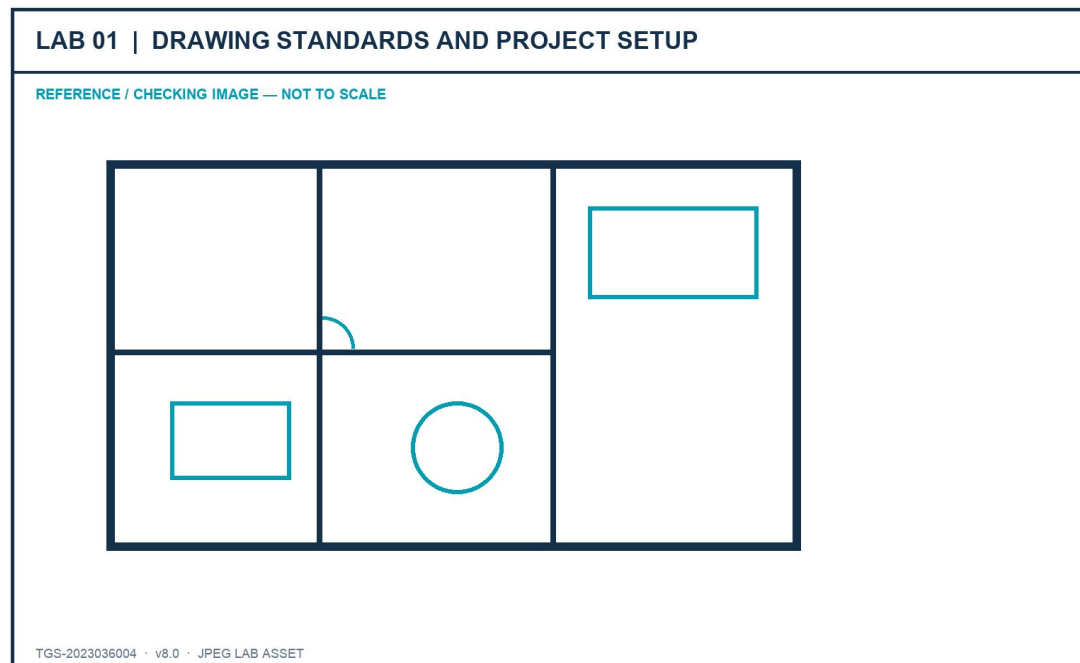
Hide / Isolate

Reduce visual complexity temporarily.

Drawing Health Maintenance



Lab 01 — Drawing Standards and Project Setup



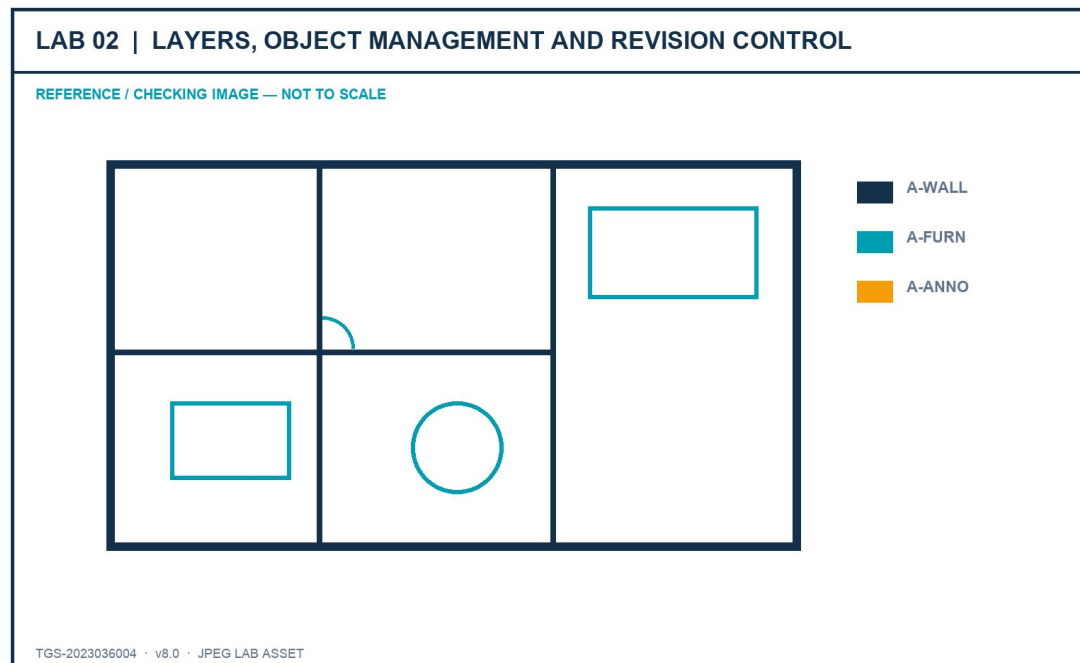
JPEG brief supplied in the individual lab folder

DECISIONS TO PROVE

- Declare units, origin and standards before editing.
- Use measurable checks instead of visual guesswork.
- Submit setup and drawing-health evidence.

Outcome: A standards-ready furniture plan with a completed drawing-standard record and verified setup evidence.

Lab 02 — Layers, Object Management and Revision Control



JPEG brief supplied in the individual lab folder

DECISIONS TO PROVE

- Organise complexity with filters and layer states.
- Use overrides without damaging global standards.
- Record controlled revision evidence.

Outcome: A revised furniture plan with reusable layer states, filters and a change log.

DOMAIN 02

Design Annotation and Detailing

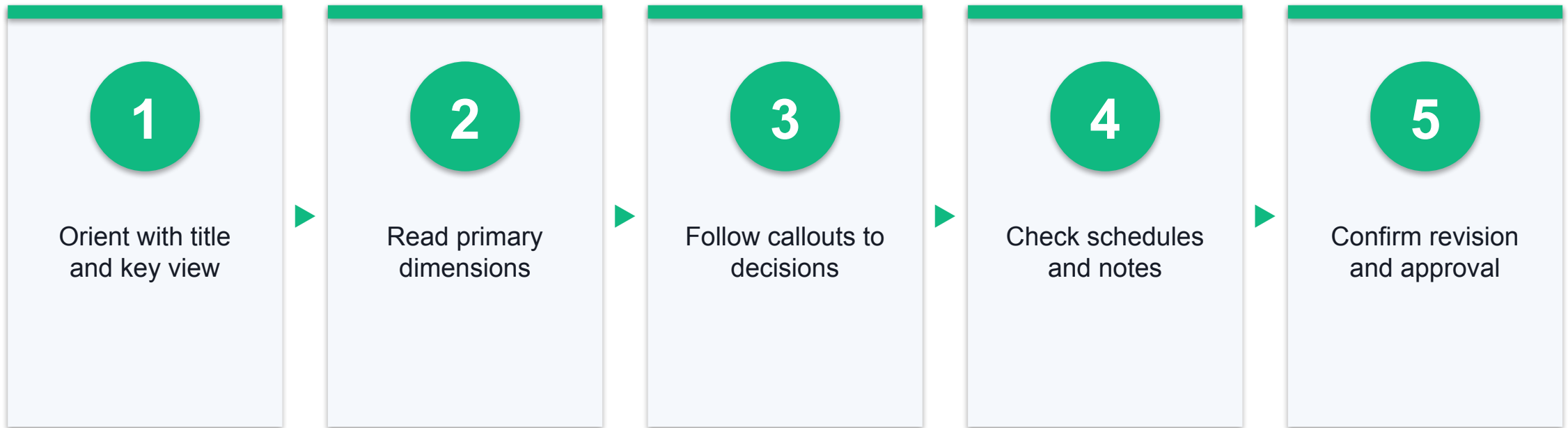
Text · dimensions · hatches · multileaders · tables · annotative scale

DESIGN COMMUNICATION

Annotation is the voice of the drawing.

A good sheet controls what the stakeholder notices, understands and decides next.

Stakeholder Reading Order



The Annotation Style System

1

Text style

Font, height policy, width and readability.

2

Dimension style

Units, precision, symbols and placement.

3

Multileader style

Arrow, landing and content hierarchy.

4

Table style

Title, header, data rows and borders.

5

Hatch settings

Pattern, scale, angle and origin.

6

Annotative properties

One object, multiple paper-readable scales.

Choose the Dimension by Intent

1

Linear / aligned

Horizontal, vertical or true-length distances.

2

Angular

The relationship between lines or features.

3

Radial / diameter

Circular geometry and manufacturing intent.

4

Ordinate

Coordinate-based production control.

5

Arc length

Measured curved distance.

6

Quick Dimension

Efficient chains that still require review.

Associativity and Overrides

Keep data connected

- Associative dimensions update with geometry.
- Reassociate after controlled geometry changes.
- Use fields when text should reflect a live property.

Control exceptions

- Avoid typing over measured values.
- Treat exceptions as documented deviations.
- Check every override in plot preview.

Hatches, Revision Clouds and Wipeouts

1

Hatch

Communicate material, section or scope — not decoration.

2

Origin

Align patterns so similar surfaces read consistently.

3

Revision cloud

Make the changed area visible and bounded.

4

Delta / note

State what changed and why it matters.

5

Wipeout

Improve local legibility without concealing required information.

6

Compare

Confirm added, removed and modified content.

Annotative Scale

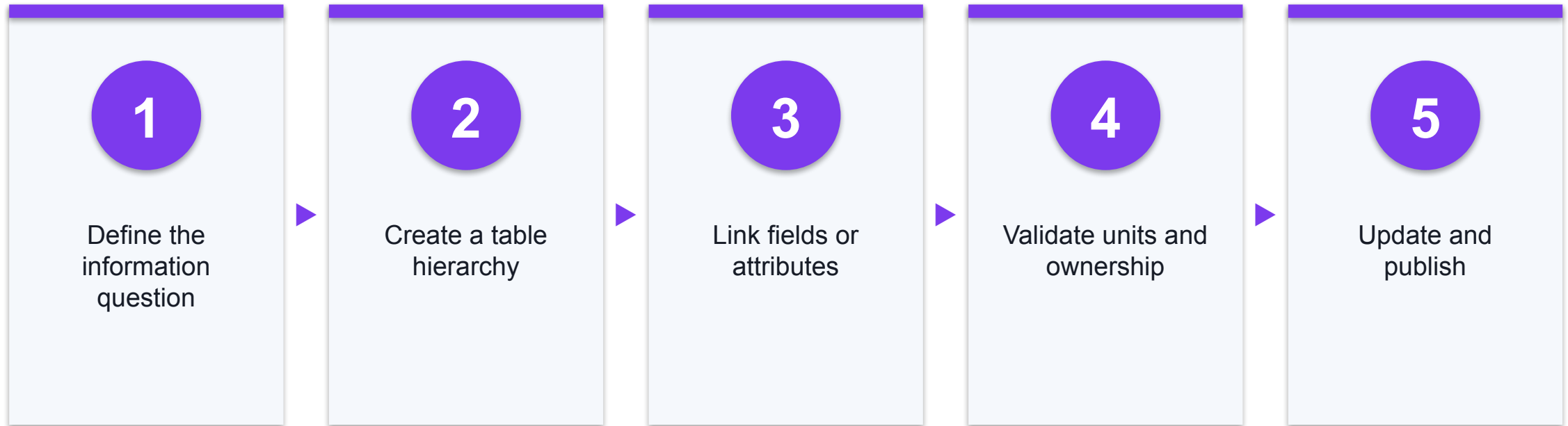
How it works

- One text or symbol can support several intended viewport scales.
- The active annotation scale controls what you create and see.
- Add only scales required by the drawing set.

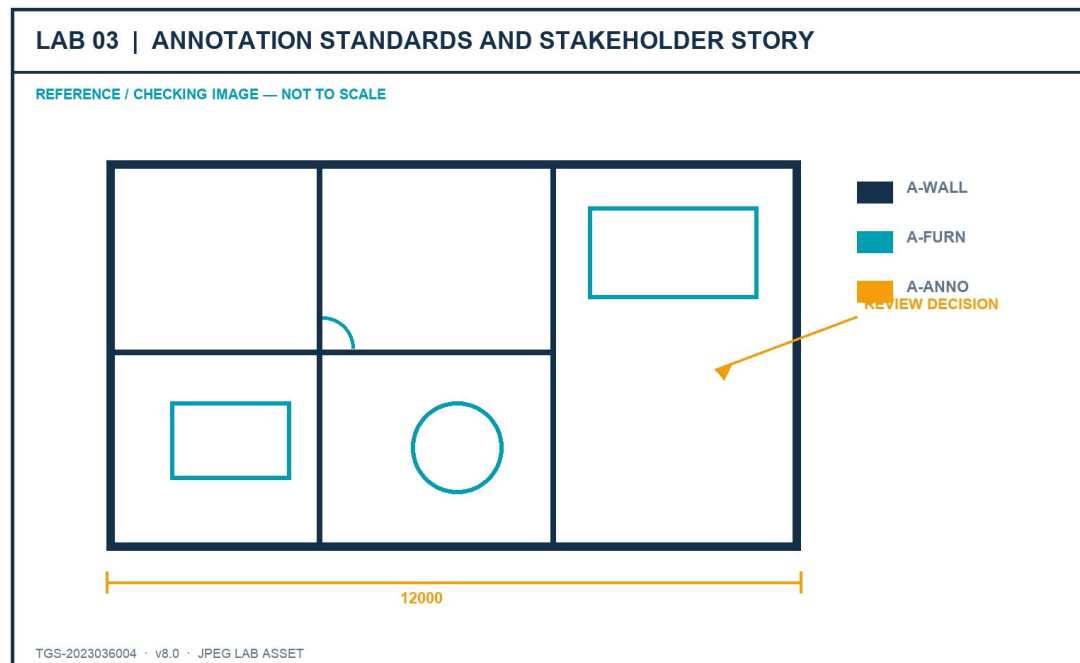
Quality checks

- Review in paper space at the target scale.
- Check viewport annotation visibility.
- Avoid duplicate manual copies with inconsistent text.

Tables and Drawing Data



Lab 03 — Annotation Standards and Stakeholder Story



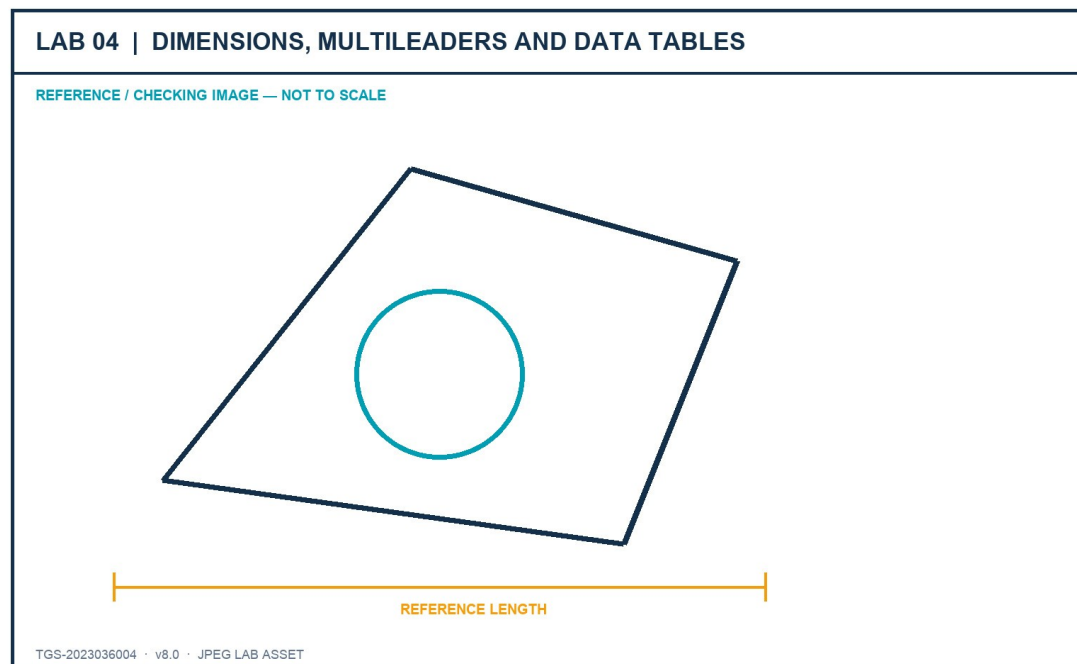
JPEG brief supplied in the individual lab folder

DECISIONS TO PROVE

- Create reusable annotation styles.
- Lead the stakeholder through a deliberate reading order.
- Make one revision visible and reviewable.

Outcome: An annotated plan with text, dimensions, multileaders, hatches and revision communication governed by named styles.

Lab 04 — Dimensions, Multileaders and Data Tables



JPEG brief supplied in the individual lab folder

DECISIONS TO PROVE

- Use fit-for-purpose dimension types.
- Coordinate leaders and component schedules.
- Prove readability in plot preview.

Outcome: A review sheet with validated dimensions, labels and a structured furniture/component table.

DOMAIN 03

Author and Edit Drawing Content

Advanced geometry · precision editing · blocks · attributes · reuse

AUTHORING WITH INTENT

Object type is a design decision.

Choose geometry by how it must behave when selected, measured, revised and reused.

Advanced Drawing Objects

1

Polyline

Connected line/arc segments as one editable object.

2

Spline

Smooth fit or control-vertex curve for refined profiles.

3

Polygon

Regular closed polyline from centre/radius or edge.

4

Region

Closed 2D area with properties and Boolean operations.

5

Xline / Ray

Infinite or semi-infinite construction reference.

6

Multiline

Parallel elements governed by a named style.

Polylines and Splines

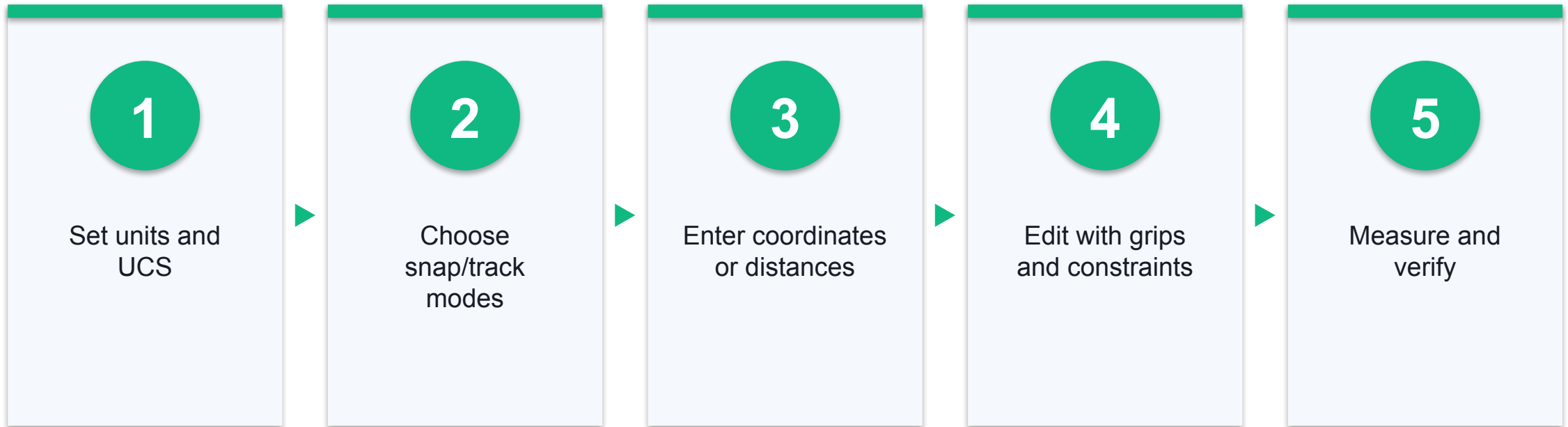
Polyline

- Polylines stay connected through grip editing.
- Width, arcs and linetype generation live on one object.
- Closed polylines support area, regions and extrusion.

Spline

- Fit splines pass through points; CV splines are shaped by a control frame.
- Use fewer, deliberate control points.
- Choose based on revision control, not smooth appearance alone.

Precision Stack



Modification Families

1

Transform

Move, Copy, Rotate, Mirror, Scale and Stretch.

2

Parallel / boundary

Offset, Trim, Extend, Break and Join.

3

Corner

Fillet and Chamfer with deliberate radii/distances.

4

Pattern

Rectangular, polar and path arrays.

5

Structure

Explode only when object structure must change.

6

Grips

Direct manipulation with object-specific options.

Associative Arrays and Multifunctional Grips

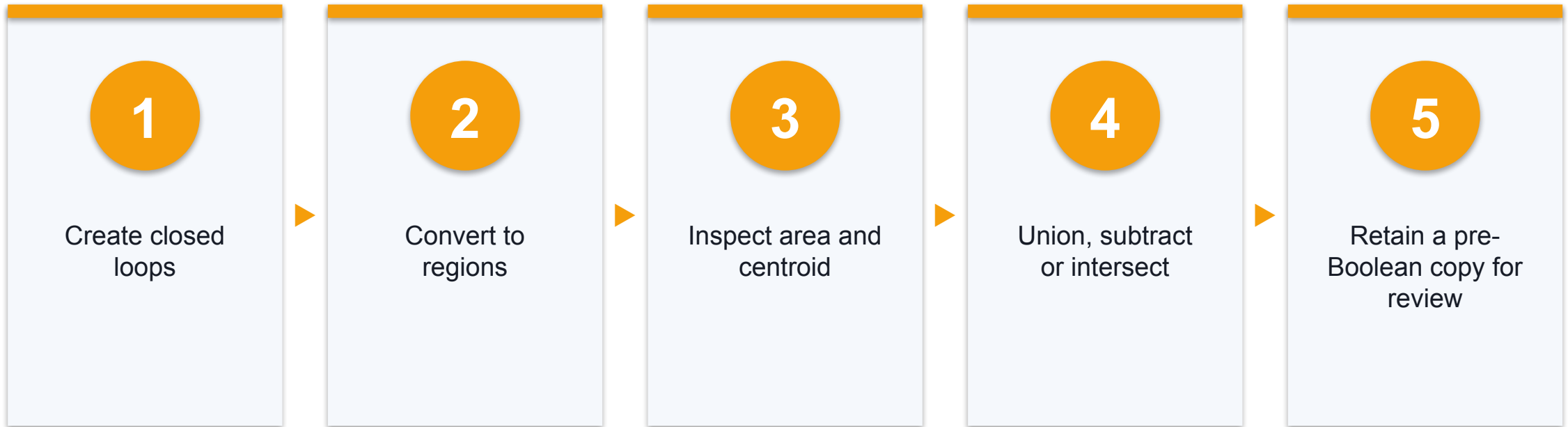
Associative arrays

- Arrays retain count, spacing, angle or path relationships.
- Edit the array as one parametric collection.
- Explode only for a documented downstream need.

Multifunctional grips

- Grip menus expose object-specific actions.
- Hover to identify Stretch, Add, Remove, Convert and other modes.
- Dynamic input shows values near the cursor.

Regions and Boolean Operations



2D Isometric Drafting

1

Isoplane

Left, Top or Right controls the active snap plane.

2

Cycle

F5 or Ctrl+E changes isoplane.

3

Isocircle

Ellipse option available only in isometric drafting.

4

Limitation

An isometric drawing is 2D representation, not a 3D model.

5

Dimensioning

Use aligned/oblique techniques deliberately.

6

Communication

Apply when a fast pictorial view supports the audience.

Blocks vs Groups

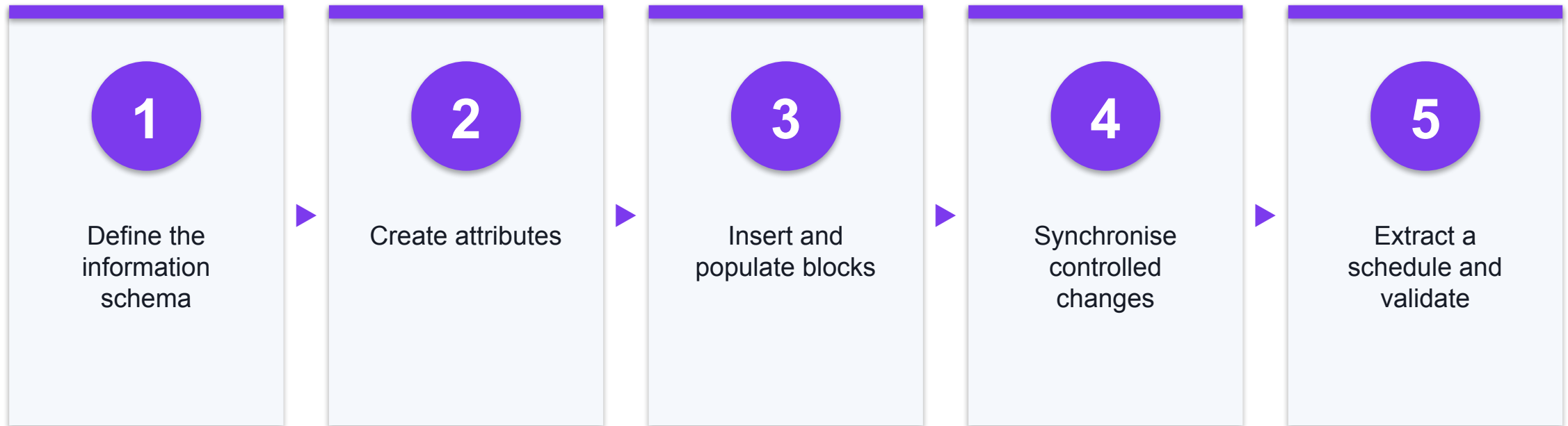
Block

- A block is one reusable definition with many instances.
- Edit the definition to update its instances.
- Attributes carry instance-specific data.

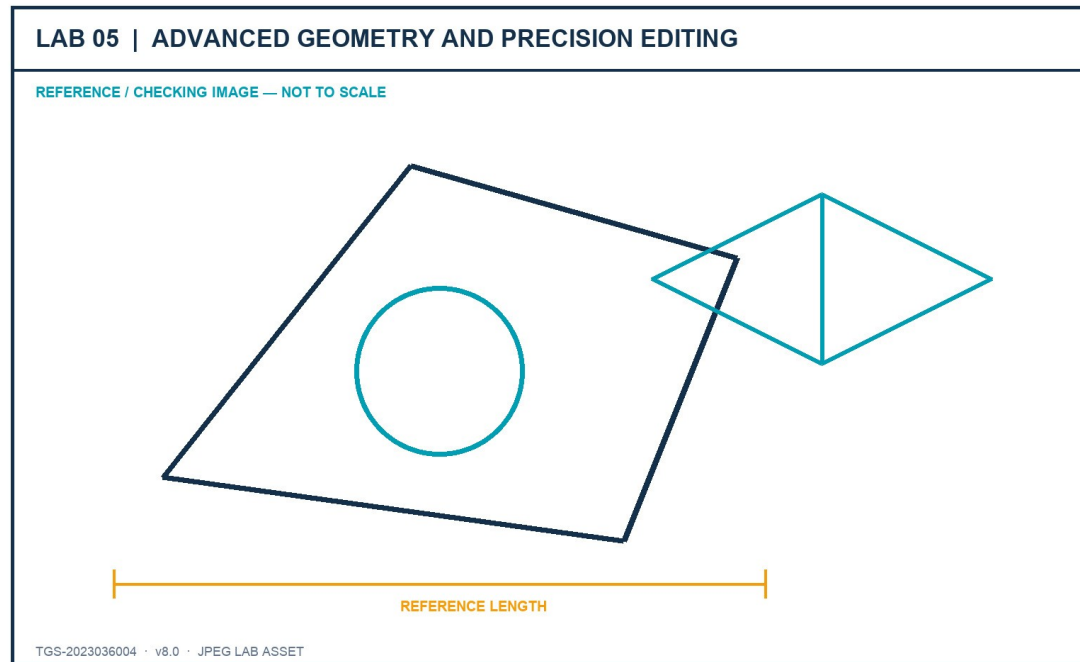
Group

- A group is a temporary selection relationship between existing objects.
- It does not create a reusable definition.
- Use it for coordinated selection, not standard content.

Attributes to Data Extraction



Lab 05 — Advanced Geometry and Precision Editing



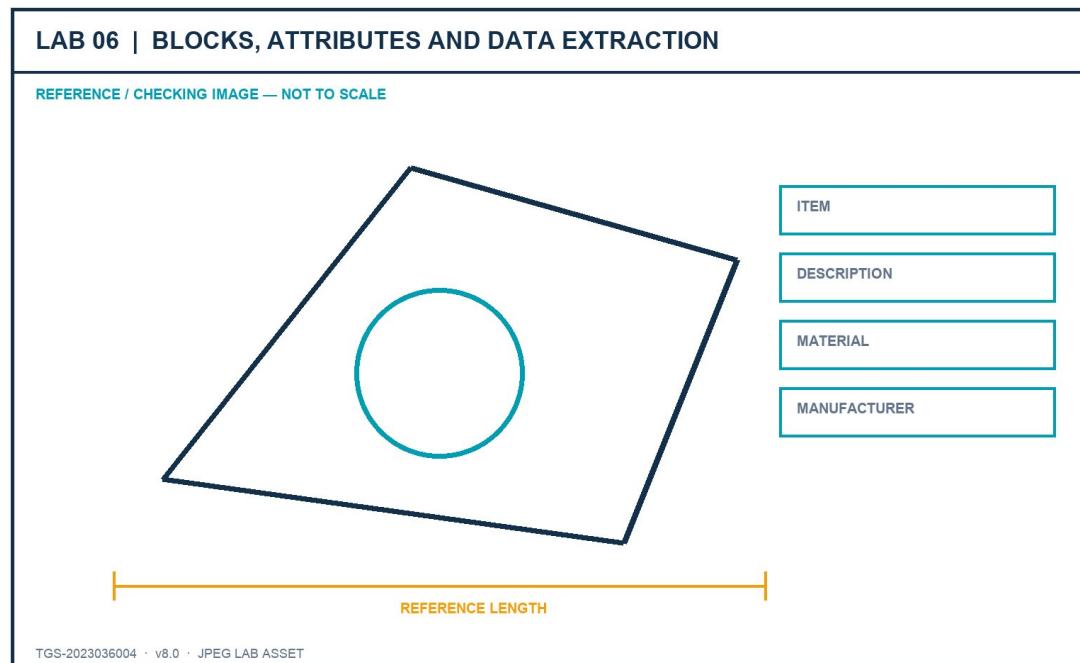
JPEG brief supplied in the individual lab folder

DECISIONS TO PROVE

- Use object types that suit revision behaviour.
- Combine precision tools into a repeatable system.
- Verify the final 2D and isometric geometry.

Outcome: A precise 2D component study plus an isometric view using fit-for-purpose object types.

Lab 06 — Blocks, Attributes and Data Extraction



JPEG brief supplied in the individual lab folder

DECISIONS TO PROVE

- Design reusable block definitions.
- Carry instance data through attributes.
- Extract and validate a component schedule.

Outcome: A small furniture block library with attributes and an extracted component schedule.

DOMAIN 04

Configure and Manage Design Output

Layouts · viewports · page setup · plotting · publishing · eTransmit

SHEET GOVERNANCE

Model once. Communicate through controlled sheets.

Paper space governs view, scale, title block and output — without duplicating model geometry.

Model Space vs Paper Space

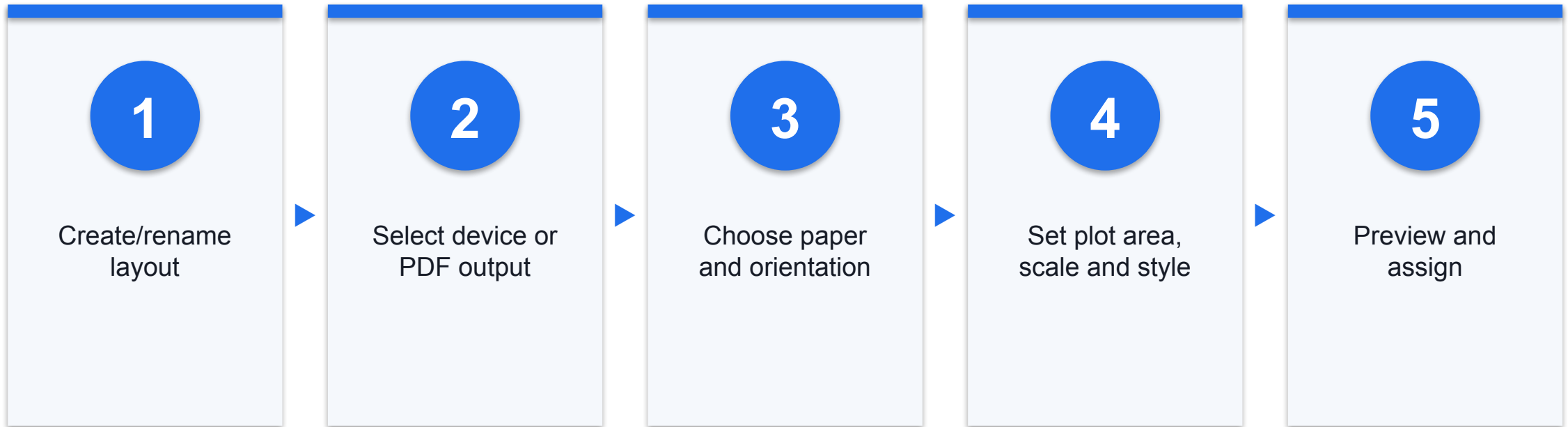
Model space

- Model at full size in agreed drawing units.
- Create and revise design geometry once.
- Maintain discipline layers and object properties.

Paper space

- Compose sheets at paper size.
- Control views through scaled viewports.
- Apply title blocks, page setups and sheet-specific emphasis.

Layout and Page Setup



Viewport Control

1

View

Top, Front, Right, Isometric or a named view.

2

Scale

Apply an approved custom or standard scale.

3

Visual style

2D wireframe, Hidden, Shaded or Realistic by purpose.

4

Layer override

Emphasise the sheet without changing the model.

5

Lock

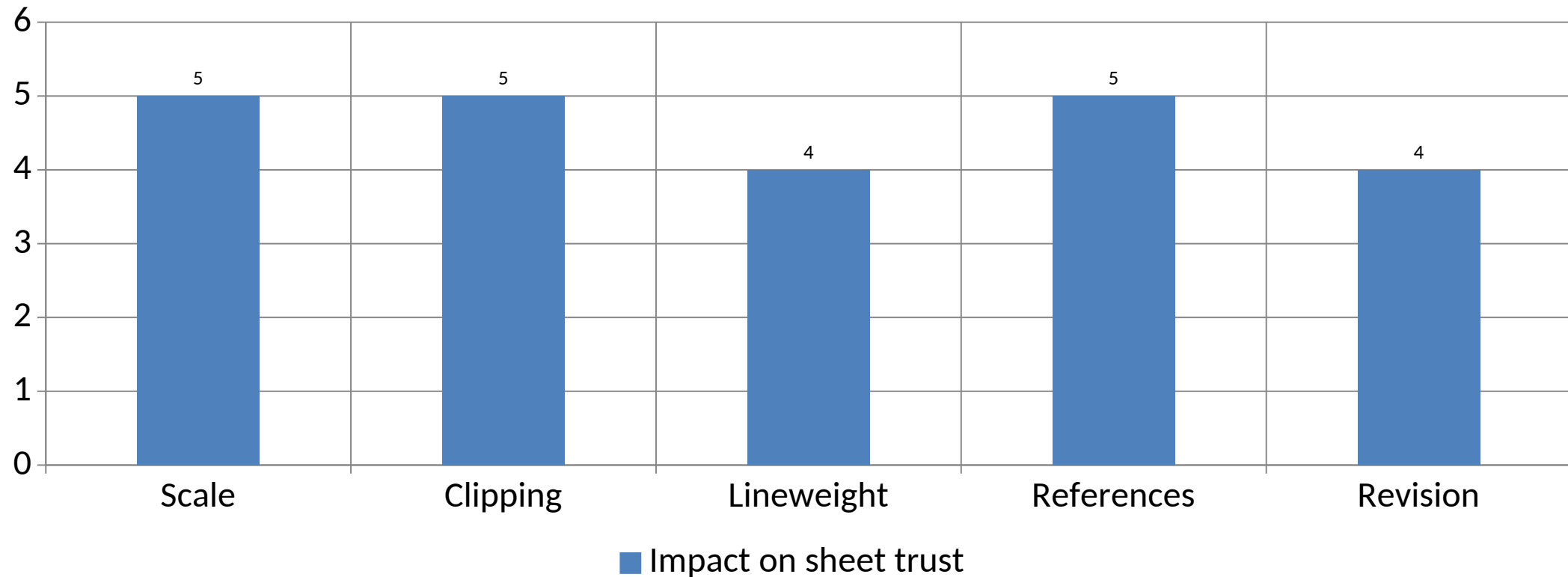
Prevent accidental pan/zoom after approval.

6

Annotation

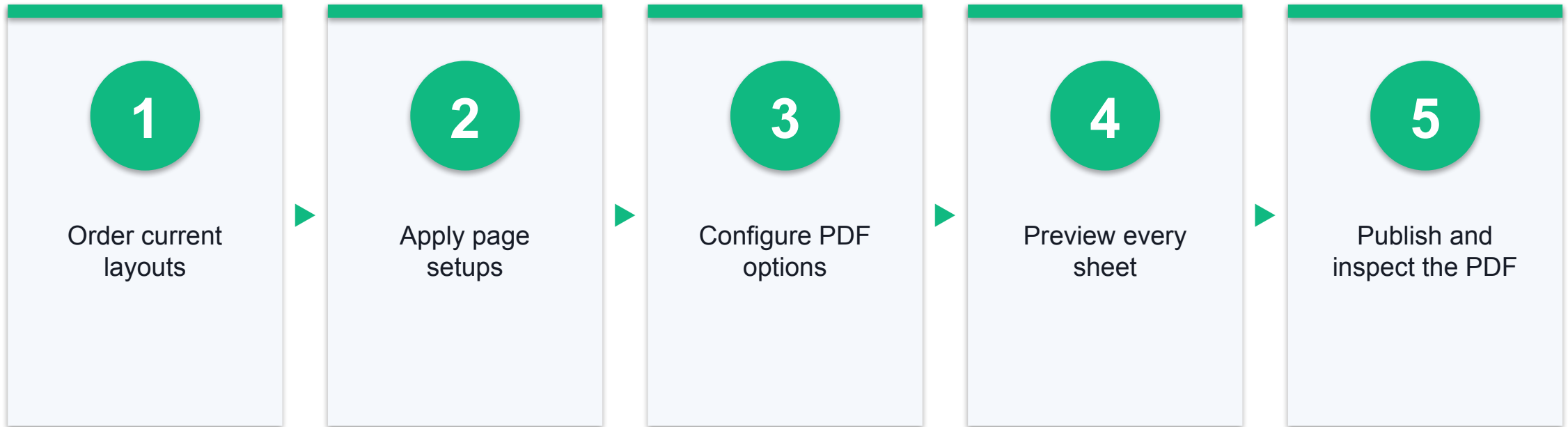
Confirm readability at the viewport scale.

Output Quality Priorities

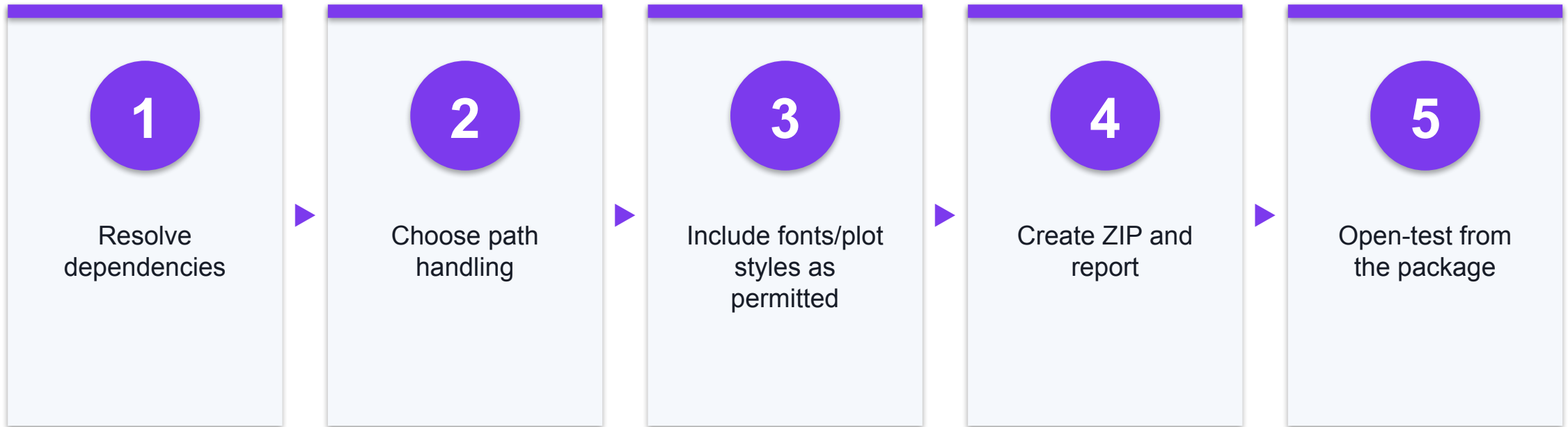


Conceptual 1–5 quality-impact comparison for discussion; values are instructional, not measured project data.

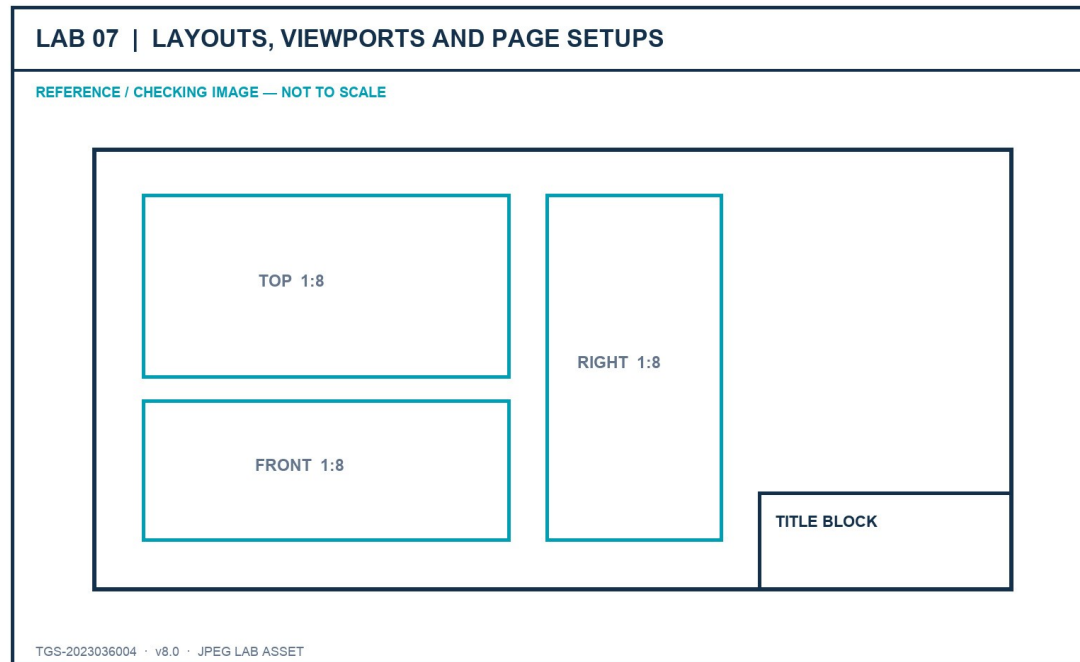
Publish a Drawing Set



eTransmit a Portable Package



Lab 07 — Layouts, Viewports and Page Setups



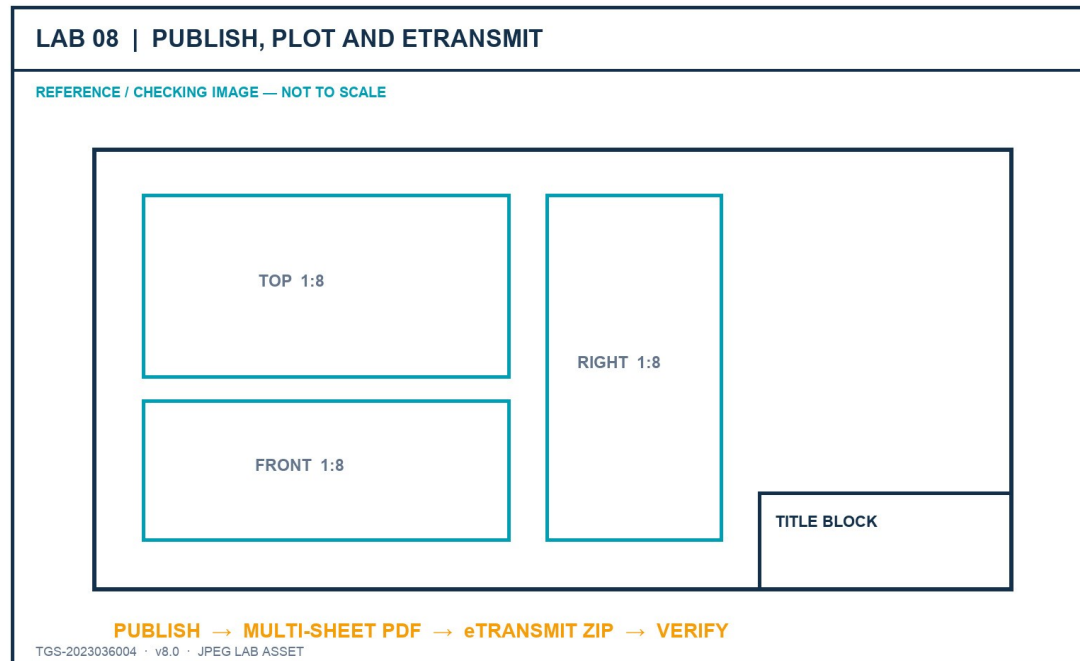
JPEG brief supplied in the individual lab folder

DECISIONS TO PROVE

- Create ISO A1, A2 and A3 sheets.
- Control views, scales, visual styles and locks.
- Prove title-block and page-setup consistency.

Outcome: A coordinated three-sheet sink drawing set that mirrors the original practical-assessment workflow.

Lab 08 — Publish, Plot and eTransmit



JPEG brief supplied in the individual lab folder

DECISIONS TO PROVE

- Publish only current, ordered layouts.
- Inspect each PDF sheet at full size.
- Verify the eTransmit inventory and portability.

Outcome: A published PDF sheet set and a verified eTransmit package with a transmittal report.

DOMAIN 05

Collaboration and Emerging Workflows

Xrefs · underlays · compare · shared views · tool palettes · automation

COORDINATION

Collaboration works when ownership stays visible.

Reference other teams' work, manage paths and communicate changes without silently duplicating source geometry.

Attach vs Overlay

Attach

- Attach allows nested references to propagate into the host.
- Use when downstream visibility is intended and controlled.
- Document nesting and ownership.

Overlay

- Overlay prevents the reference from propagating when the host is referenced again.
- Use to avoid circular or unwanted nesting.
- Often safer for discipline coordination.

Reference Path Strategy

1

Relative

Portable within a known project folder structure.

2

Full

Exact location but fragile when moved between machines.

3

No path

Search-based and ambiguous outside a disciplined package.

4

Repath

Repair links without changing insertion or scale.

5

Unload / reload

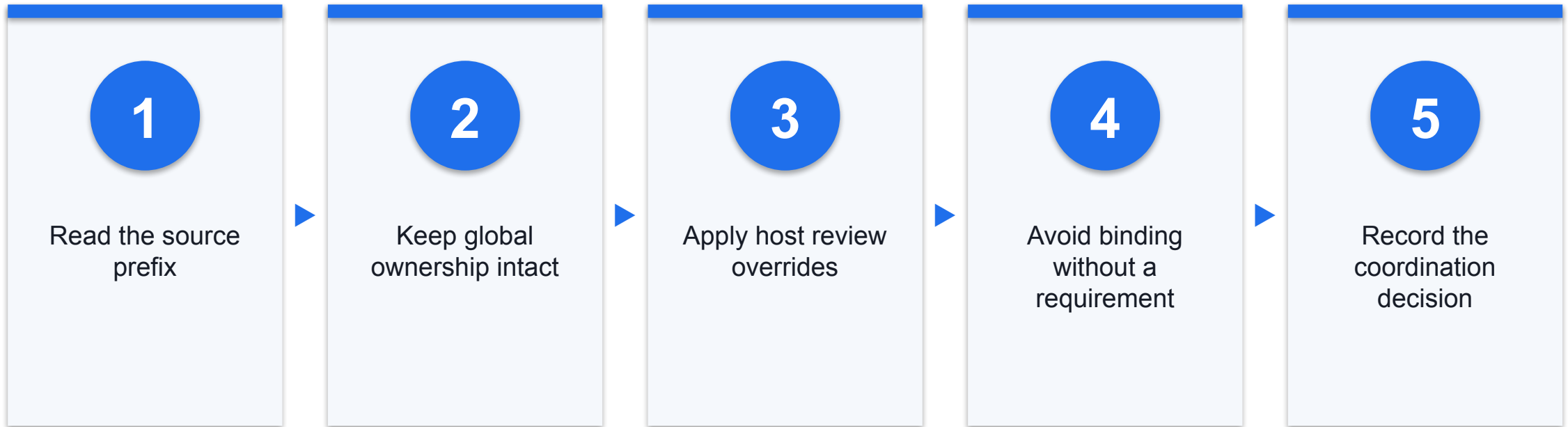
Control performance while retaining the definition.

6

eTransmit

Package references and document path handling.

Coordinate Xref Layers



Underlays and Linked Evidence

1

PDF underlay

Reference issued sheets without redrawing them.

2

Raster image

Trace, compare or communicate using controlled imagery.

3

Clip

Limit the visible region without modifying the source.

4

Fade / contrast

Keep context visible without overpowering host content.

5

Hyperlink

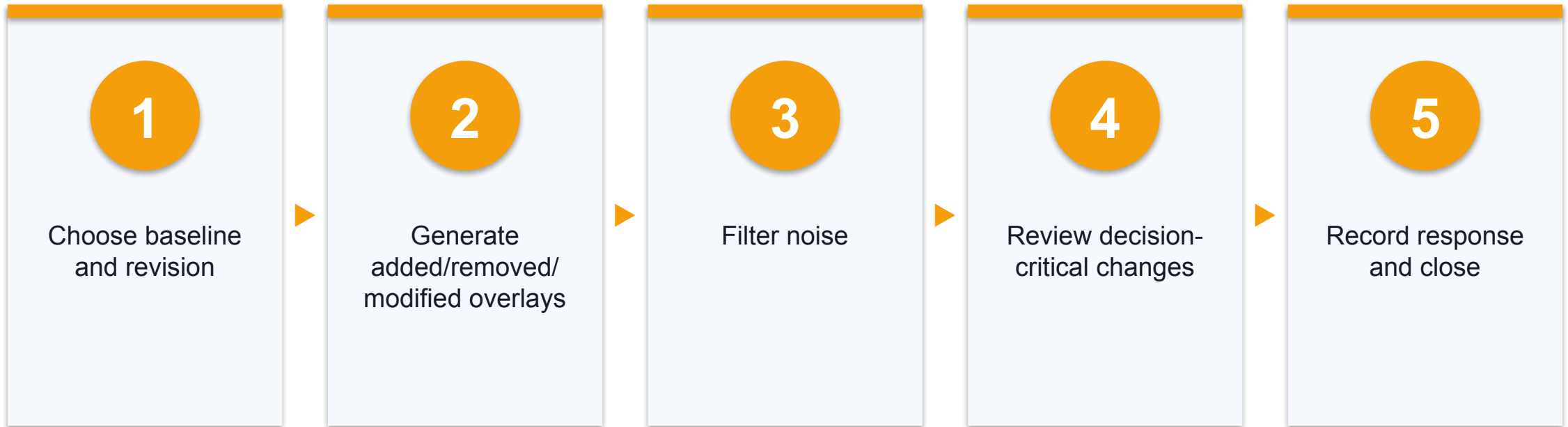
Connect an object to approved external information.

6

Path check

Verify links from the final package location.

Drawing Compare



Evaluate Emerging Workflows

1

Shared Views

Browser-based review without handing over editable source.

2

Markup Import

Bring review comments back into the drawing context.

3

Data extraction

Turn block/attribute data into schedules and QA evidence.

4

Tool palettes

Standardise approved commands, blocks and hatches.

5

Automation

Use scripts and routines for repeatable, governed tasks.

6

Cloud coordination

Assess permissions, ownership, revision and audit trail.

Technology Selection Scorecard

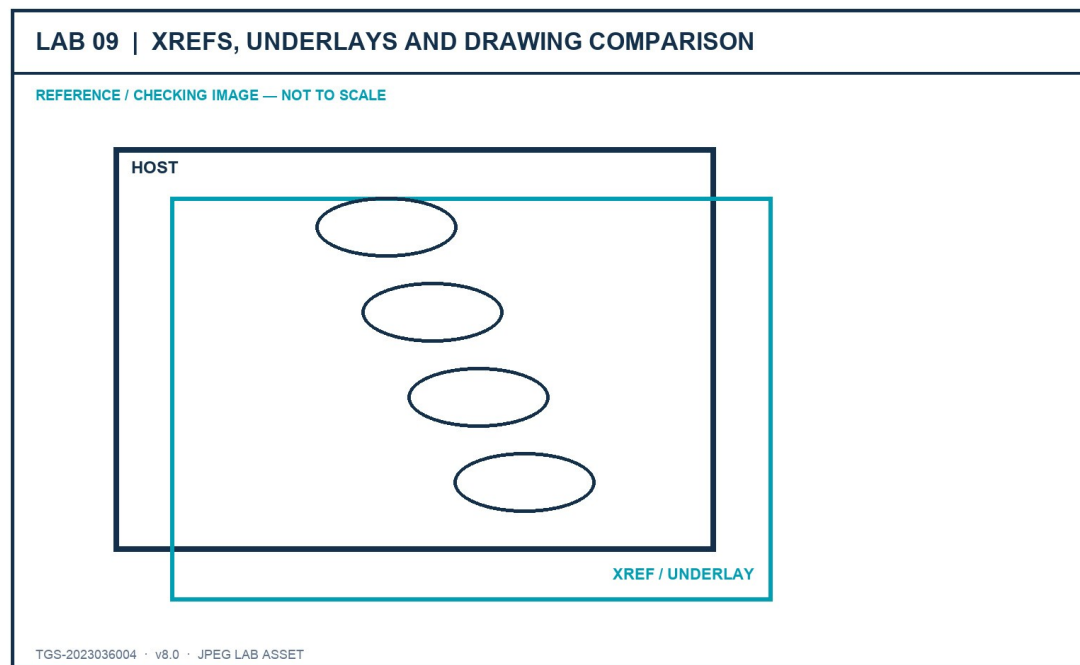
Risk and quality

- Accuracy and standards fit
- Interoperability and path portability
- Revision traceability
- Security and access control

Value and growth

- Time saved across repeated work
- Learning and support effort
- Evidence and auditability
- Ability to scale with organisational growth

Lab 09 — Xrefs, Underlays and Drawing Comparison



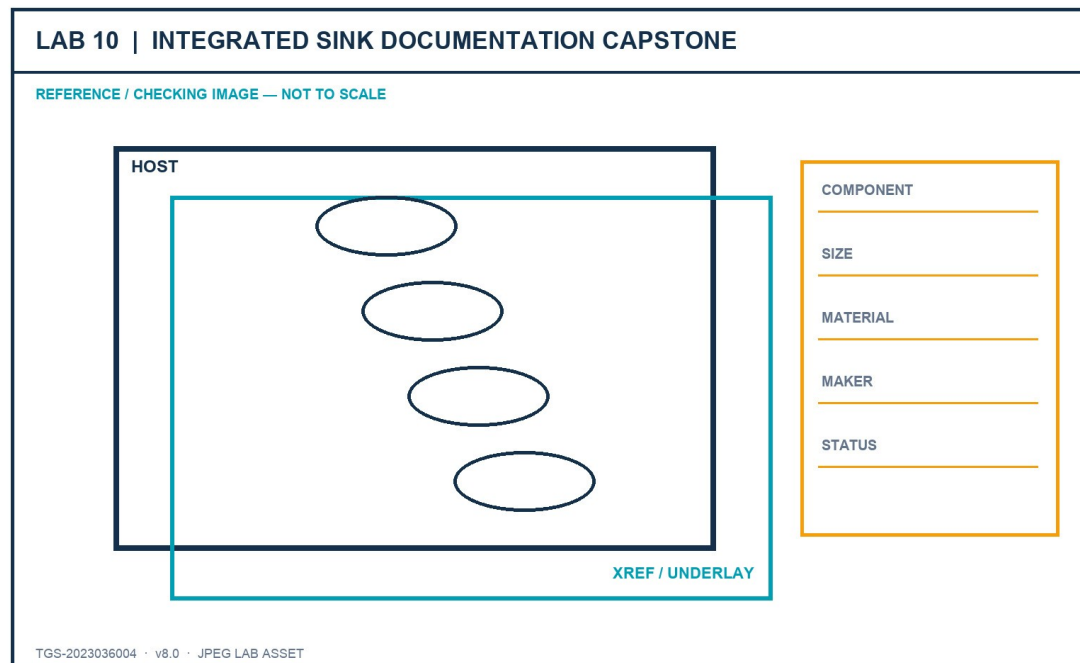
JPEG brief supplied in the individual lab folder

DECISIONS TO PROVE

- Preserve ownership through controlled references.
- Use relative paths and host-only review overrides.
- Classify changes and stakeholder response.

Outcome: A host coordination drawing with controlled Xrefs/underlays and a documented comparison review.

Lab 10 — Integrated Sink Documentation Capstone



JPEG brief supplied in the individual lab folder

DECISIONS TO PROVE

- Integrate standards, layouts and component data.
- Publish and package a complete sink drawing set.
- Lead a concise stakeholder approval story.

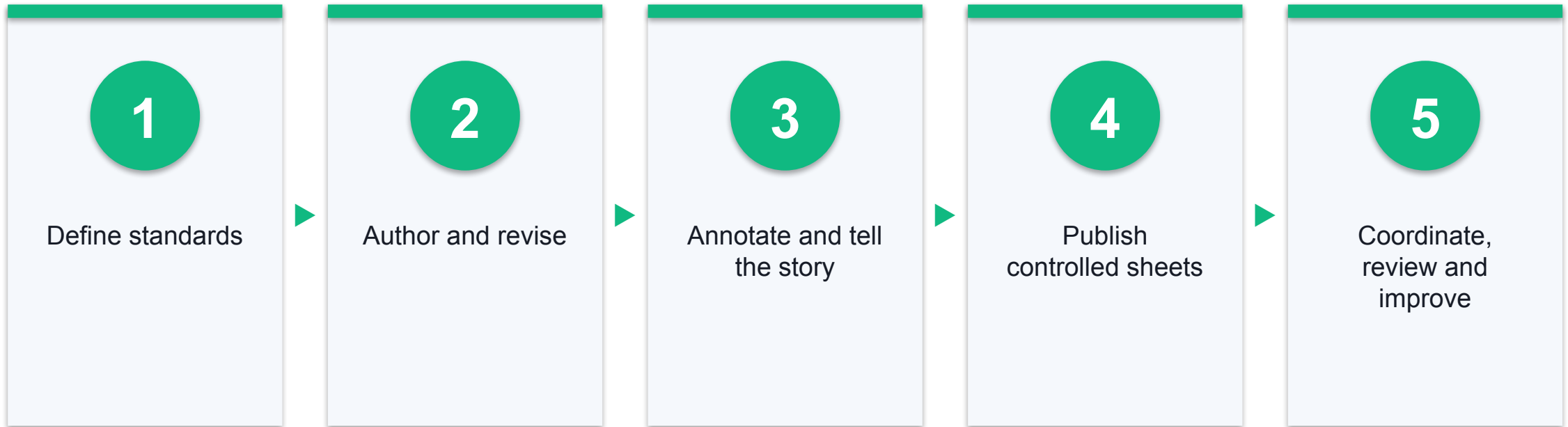
Outcome: A standards-based sink drawing set with layouts, title blocks, viewports, component table, publish package and review evidence.



WRAP-UP

Course Summary and Assessment

Integrated Professional Workflow



Evidence Quality Checklist

1

Show the setting and the visible result together.

2

Use filenames that identify lab, learner and revision.

3

Verify units, coordinates, scales and paths explicitly.

4

Retain current DWG plus published/package outputs.

5

Capture acceptance criteria, not every click.

6

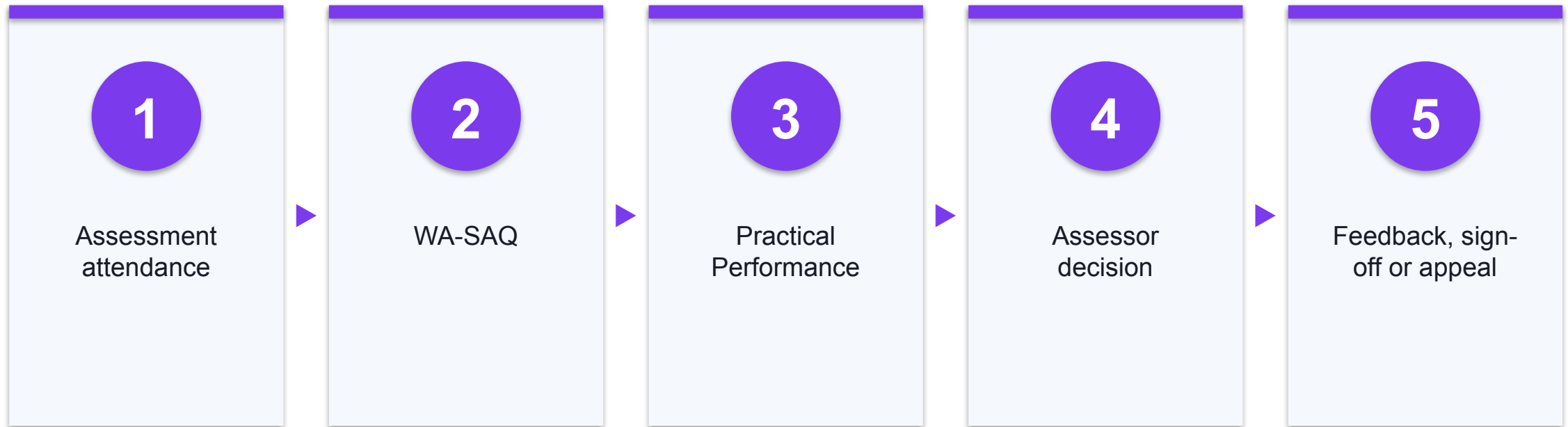
Explain the decision behind the tool choice.

Final Assessment

- Written Assessment (WA-SAQ) — 4 open-ended questions covering K1–K4, 60 minutes, open book.
- Practical Performance (PP) — 5 hands-on tasks covering A1–A5, 120 minutes, open book.
- Question papers retain the original instrument type, counts, timings and K/A coverage.
- Answer keys remain trainer-only and are never linked to the LMS or public GitHub repository.

COMPLETE EVERY STAGE

Assessment and Sign-off Flow



Digital Attendance — Final Check

1

AM

Completed for Day 4.

2

PM

Completed after lunch.

3

Assessment

Completed before the final papers.

4

TRAQOM

Submit the required feedback survey.

Support and Next Learning

Course support

- Revisit the detailed Learner Guide and individual lab folders.
- Use only files and references authorised for your project.
- Request post-course consultation through Tertiary Infotech.

Professional development

- Build a portfolio drawing set with standards evidence.
- Practise explaining design decisions without relying on memorised clicks.
- Continue with related Autodesk design and BIM workflows.

AUTOCAD DESIGN AND DRAFTING

Thank You!

Draft with intent. Revise with control. Publish with confidence.